

SOLIDWORKS API (.NET VBA C#) — Reference (271/337): ISketchManager (part 1 of 3)

SOLIDWORKS API Documentation (converted from CHM)

ISketchManager (part 1 of 3) (Type)

ISketchManager Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchManager Interface

Provides access to sketch-creation routines.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISketchManager
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
```

C#

```
public interface ISketchManager
```

C++/CLI

```
public interface class ISketchManager
```

Visual Basic for Applications (VBA) Syntax

See [SketchManager](#).

Example

[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)
[Extend Sketch Entity \(VBA\)](#)
[Offset Selected Edges in Active Sketch \(VBA\)](#)
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[Create Block Definition and Insert Block Instance \(C#\)](#)
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[Create Block Definition and Insert Block Instance \(VB.NET\)](#)

Accessors

[IModelDoc2::SketchManager](#)

Access Diagram

[SketchManager](#)

See Also

[ISketchManager Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISketchManager Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchManager Interface

For a list of all members of this type, see [ISketchManager members](#).

Public Methods

Name	Description
 AddAlongXDimension	Projects and displays along the x axis a dimension between selected points in a 3D sketch.
 AddAlongYDimension	Projects and displays along the y axis a dimension between selected points in a 3D sketch.
 AddAlongZDimension	Projects and displays along the z axis a dimension between selected points in a 3D sketch.
 ConvertEntities	Not implemented. Use ISketchManager::SketchUseEdge2
 Create3PointArc	Creates a 3-point arc.
 Create3PointCenterRectangle	Creates a 3-point center rectangle at any angle.
 Create3PointCornerRectangle	Creates a 3-point corner rectangle at any angle.
 CreateArc	Creates an arc based on a center point, a start point, an end point, and a direction.
 CreateBoundaryHatch	Creates area hatch/fill boundary hatches using closed sketch profiles.
 CreateCenterLine	Creates a center line between the specified points.
 CreateCenterRectangle	Creates a center rectangle.
 CreateChamfer	Creates a chamfer between two selected sketch entities.
 CreateCircle	Creates a circle based on a center point and a point on the circle.
 CreateCircleByRadius	Creates a circle based on a center point and a specified radius.
 CreateCircularSketchStepAndRepeat	Creates circular sketch pattern.
 CreateConic	Creates a conic curve in the active sketch.
 CreateConstructionGeometry	Sets selected sketch segments to be construction geometry instead of sketch geometry.
 CreateCornerRectangle	Creates a corner rectangle.
 CreateEllipse	Creates an ellipse using the specified center, major-axis, and minor-axis points.
 CreateEllipticalArc	Creates a partial ellipse given a center point, two points that specify the major and minor axis, and two points that define the elliptical start and end points.
 CreateEquationSpline	Obsolete. Superseded by ISketchManager::CreateEquationSpline2
 CreateEquationSpline2	Creates an equation-driven 2D explicit or parametric curve or a 3D parametric curve.
 CreateFillet	Creates a sketch fillet using the selected sketch entities.
 CreateLine	Creates a sketch line in the currently active 2D or 3D sketch.
 CreateLinearSketchStepAndRepeat	Creates a linear sketch pattern.
 CreateParabola	Creates a parabola in the active sketch.
 CreateParallelogram	Creates a parallelogram.
 CreatePoint	Creates a sketch point in the active 2D or 3D sketch.
 CreatePolygon	Creates a polygon in the active sketch.
 CreateRegionHatch	Creates an area hatch/fill region hatch using a closed sketch profile.
 CreateSketchBelt	Creates a sketch belt feature.
 CreateSketchPlane	Creates a 3D sketch plane.
 CreateSketchSlot	Creates a sketch slot.
 CreateSpline	Obsolete. Superseded by ISketchManager::CreateSpline2
 CreateSpline2	Obsolete. Superseded by ISketchManager::CreateSpline3
 CreateSpline3	Creates either a 2D spline or a spline constrained to a surface.
 CreateSplineByEqnParams	Creates a B-curve from B-spline data; that is, a set of B-spline vertices (control points) and a knot vector.
 CreateSplineParamData	Creates an empty spline parameter data object.
 CreateSplinesByEqnParams	Obsolete. Superseded by ISketchManager::CreateSplinesByEqnParams2
 CreateSplinesByEqnParams2	Creates one or more spline segments using the B-curve parameters provided.
 CreateTangentArc	Creates a tangent arc.
 EditCircularSketchStepAndRepeat	Edits a circular sketch pattern.

	EditLinearSketchStepAndRepeat	Edits a linear sketch pattern.
	EditSketchBlock	Puts the block definition in edit mode.
	EndEditSketchBlock	Saves or discards your edits of the block and then ends the current editing session of this block.
	ExplodeSketchBlockInstance	Explodes the specified block instance.
	FullyDefineSketch	Fully defines a sketch.
	GetDynamicMirror	Gets whether dynamic sketch mirroring, which is the automatic mirroring of newly created sketch entities about a selected centerline, is enabled.
	GetSketchBlockDefinitionCount	Gets the number of block definitions in the model.
	GetSketchBlockDefinitions	Gets all of the block definitions.
	ICreateSpline	Obsolete. Superseded by ISketchManager.ICreateSpline2
	ICreateSpline2	Creates a spline passing through the given points.
	ICreateSplineByEqnParams	Creates a B-curve from B-spline data; that is, a set of B-spline vertices (control points) and a knot vector.
	ICreateSplinesByEqnParams	Creates one or more spline segments using the B-curve parameters provided.
	IGetSketchBlockDefinitions	Gets all of the block definitions.
	Insert3DSketch	Inserts a new 3D sketch in a model or closes the active sketch.
	InsertExplodeLineSketch	Inserts or closes an explode line sketch in an exploded view.
	InsertSketch	Inserts a new sketch in the current part or assembly document.
	InsertSketchBlockInstance	Inserts a block instance at the specified location using the block definition.
	InsertSketchPicture	Obsolete. Superseded by ISketchManager.InsertSketchPicture2
	InsertSketchPicture2	Inserts a picture on the current drawing sketch.
	IntersectCurves	Creates a sketched intersection curve.
	MakeSketchBlockFromFile	Creates a block definition using the specified file.
	MakeSketchBlockFromSelected	Creates a block definition at the specified location from the selected entities.
	MakeSketchBlockFromSketch	Creates a block definition at the specified location using all of the sketch entities in the active sketch.
	MakeSketchChain	Creates a sketch path using the selected entities.
	PerimeterCircle	Draws a 3-point perimeter arc.
	ReverseEndPointTangent	Reverses the end point tangent direction of splines and arcs.
	RotateOrCopy3DAboutVector	Rotates, and optionally copies, the selected 3D sketch entities about the specified vector.
	RotateOrCopy3DAboutXYZ	Rotates, and optionally copies, the selected 3D sketch entities about the specified x, y, and z coordinates.
	SetDynamicMirror	Enables or disables dynamic sketch mirroring, which is the automatic mirroring of newly created sketch entities about a selected centerline.
	SetGridOptions	Sets the options for the grid.
	SketchExtend	Adds to the length of the selected sketch entity (i.e., line, centerline, or arc) to meet the nearest sketch entity.
	SketchOffset	Obsolete. Superseded by ISketchManager.SketchOffset2 .
	SketchOffset2	Offsets the selected sketch entities.
	SketchReplace	Obsolete. Superseded by ISketchManager.SketchReplace2
	SketchReplace2	Replaces a sketch entity in a model with another sketch entity, preserving all references.
	SketchTrim	Trims the selected sketch entities.
	SketchUseEdge	Obsolete. Superseded by ISketchManager.SketchUseEdge2
	SketchUseEdge2	Obsolete. Superseded by ISketchManager.SketchUseEdge3
	SketchUseEdge3	Creates sketch entities on a sketch plane by projecting selected edges, loops, faces, curves, and external sketch contours.
	SplitClosedSegment	Splits the selected closed sketch segment into two sketch segments.
	SplitOpenSegment	Splits the selected open sketch segment into two sketch segments.

See Also

[ISketchManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISketchManager Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISketchManager Interface

The following tables list the members exposed by [ISketchManager](#).

Public Properties

Name	Description
 ActiveSketch	Gets the active sketch.
 AddToDB	Gets or sets whether sketch entities are added directly to the SOLIDWORKS database.
 AutoInference	Obsolete. Superseded by ISlidWorks::GetUserPreferenceToggle or ISlidWorks::SetUserPreferenceToggle and swUserPreferenceToggle_e_swSketchInference .
 AutoSolve	Gets or sets whether SOLIDWORKS automatically solves the sketch geometry of the part while creating it.
 CurvatureDensity	Gets or sets the scaling factor by which to adjust the density of the curvature combs for this spline.
 CurvatureScale	Gets or sets the scaling factor by which to adjust the size of the curvature combs for this spline.
 DisplayWhenAdded	Gets or sets whether new sketch entities are immediately displayed when created.
 Document	Gets the document for this ISketchManager object.
 InferenceMode	Obsolete. Superseded by ISlidWorks::GetUserPreferenceToggle or ISlidWorks::SetUserPreferenceToggle and swUserPreferenceToggle_e_swSketchInference .

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Public Methods

Name	Description
 AddAlongXDimension	Projects and displays along the x axis a dimension between selected points in a 3D sketch.
 AddAlongYDimension	Projects and displays along the y axis a dimension between selected points in a 3D sketch.
 AddAlongZDimension	Projects and displays along the z axis a dimension between selected points in a 3D sketch.
 ConvertEntities	Not implemented. Use ISketchManager::SketchUseEdge2
 Create3PointArc	Creates a 3-point arc.
 Create3PointCenterRectangle	Creates a 3-point center rectangle at any angle.
 Create3PointCornerRectangle	Creates a 3-point corner rectangle at any angle.
 CreateArc	Creates an arc based on a center point, a start point, an end point, and a direction.
 CreateBoundaryHatch	Creates area hatch/fill boundary hatches using closed sketch profiles.
 CreateCenterLine	Creates a center line between the specified points.
 CreateCenterRectangle	Creates a center rectangle.
 CreateChamfer	Creates a chamfer between two selected sketch entities.
 CreateCircle	Creates a circle based on a center point and a point on the circle.
 CreateCircleByRadius	Creates a circle based on a center point and a specified radius.
 CreateCircularSketchStepAndRepeat	Creates circular sketch pattern.
 CreateConic	Creates a conic curve in the active sketch.
 CreateConstructionGeometry	Sets selected sketch segments to be construction geometry instead of sketch geometry.
 CreateCornerRectangle	Creates a corner rectangle.
 CreateEllipse	Creates an ellipse using the specified center, major-axis, and minor-axis points.
 CreateEllipticalArc	Creates a partial ellipse given a center point, two points that specify the major and minor axis, and two points that define the elliptical start and end points.

	CreateEquationSpline	Obsolete. Superseded by ISketchManager::CreateEquationSpline2
	CreateEquationSpline2	Creates an equation-driven 2D explicit or parametric curve or a 3D parametric curve.
	CreateFillet	Creates a sketch fillet using the selected sketch entities.
	CreateLine	Creates a sketch line in the currently active 2D or 3D sketch.
	CreateLinearSketchStepAndRepeat	Creates a linear sketch pattern.
	CreateParabola	Creates a parabola in the active sketch.
	CreateParallelogram	Creates a parallelogram.
	CreatePoint	Creates a sketch point in the active 2D or 3D sketch.
	CreatePolygon	Creates a polygon in the active sketch.
	CreateRegionHatch	Creates an area hatch/fill region hatch using a closed sketch profile.
	CreateSketchBelt	Creates a sketch belt feature.
	CreateSketchPlane	Creates a 3D sketch plane.
	CreateSketchSlot	Creates a sketch slot.
	CreateSpline	Obsolete. Superseded by ISketchManager::CreateSpline2
	CreateSpline2	Obsolete. Superseded by ISketchManager::CreateSpline3
	CreateSpline3	Creates either a 2D spline or a spline constrained to a surface.
	CreateSplineByEqnParams	Creates a B-curve from B-spline data; that is, a set of B-spline vertices (control points) and a knot vector.
	CreateSplineParamData	Creates an empty spline parameter data object.
	CreateSplinesByEqnParams	Obsolete. Superseded by ISketchManager::CreateSplinesByEqnParams2
	CreateSplinesByEqnParams2	Creates one or more spline segments using the B-curve parameters provided.
	CreateTangentArc	Creates a tangent arc.
	EditCircularSketchStepAndRepeat	Edits a circular sketch pattern.
	EditLinearSketchStepAndRepeat	Edits a linear sketch pattern.
	EditSketchBlock	Puts the block definition in edit mode.
	EndEditSketchBlock	Saves or discards your edits of the block and then ends the current editing session of this block.
	ExplodeSketchBlockInstance	Explodes the specified block instance.
	FullyDefineSketch	Fully defines a sketch.
	GetDynamicMirror	Gets whether dynamic sketch mirroring, which is the automatic mirroring of newly created sketch entities about a selected centerline, is enabled.
	GetSketchBlockDefinitionCount	Gets the number of block definitions in the model.
	GetSketchBlockDefinitions	Gets all of the block definitions.
	ICreateSpline	Obsolete. Superseded by ISketchManager::ICreateSpline2
	ICreateSpline2	Creates a spline passing through the given points.
	ICreateSplineByEqnParams	Creates a B-curve from B-spline data; that is, a set of B-spline vertices (control points) and a knot vector.
	ICreateSplinesByEqnParams	Creates one or more spline segments using the B-curve parameters provided.
	IGetSketchBlockDefinitions	Gets all of the block definitions.
	Insert3DSketch	Inserts a new 3D sketch in a model or closes the active sketch.
	InsertExplodeLineSketch	Inserts or closes an explode line sketch in an exploded view.
	InsertSketch	Inserts a new sketch in the current part or assembly document.
	InsertSketchBlockInstance	Inserts a block instance at the specified location using the block definition.
	InsertSketchPicture	Obsolete. Superseded by ISketchManager::InsertSketchPicture2
	InsertSketchPicture2	Inserts a picture on the current drawing sketch.
	IntersectCurves	Creates a sketched intersection curve.
	MakeSketchBlockFromFile	Creates a block definition using the specified file.

		
	MakeSketchBlockFromSelected	Creates a block definition at the specified location from the selected entities.
	MakeSketchBlockFromSketch	Creates a block definition at the specified location using all of the sketch entities in the active sketch.
	MakeSketchChain	Creates a sketch path using the selected entities.
	PerimeterCircle	Draws a 3-point perimeter arc.
	ReverseEndPointTangent	Reverses the end point tangent direction of splines and arcs.
	RotateOrCopy3DAboutVector	Rotates, and optionally copies, the selected 3D sketch entities about the specified vector.
	RotateOrCopy3DAboutXYZ	Rotates, and optionally copies, the selected 3D sketch entities about the specified x, y, and z coordinates.
	SetDynamicMirror	Enables or disables dynamic sketch mirroring, which is the automatic mirroring of newly created sketch entities about a selected centerline.
	SetGridOptions	Sets the options for the grid.
	SketchExtend	Adds to the length of the selected sketch entity (i.e., line, centerline, or arc) to meet the nearest sketch entity.
	SketchOffset	Obsolete. Superseded by ISketchManager::SketchOffset2 .
	SketchOffset2	Offsets the selected sketch entities.
	SketchReplace	Obsolete. Superseded by ISketchManager::SketchReplace2 .
	SketchReplace2	Replaces a sketch entity in a model with another sketch entity, preserving all references.
	SketchTrim	Trims the selected sketch entities.
	SketchUseEdge	Obsolete. Superseded by ISketchManager::SketchUseEdge2 .
	SketchUseEdge2	Obsolete. Superseded by ISketchManager::SketchUseEdge3 .
	SketchUseEdge3	Creates sketch entities on a sketch plane by projecting selected edges, loops, faces, curves, and external sketch contours.
	SplitClosedSegment	Splits the selected closed sketch segment into two sketch segments.
	SplitOpenSegment	Splits the selected open sketch segment into two sketch segments.

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See Also

[ISketchManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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ISketchManager Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchManager Interface

For a list of all members of this type, see [ISketchManager members](#).

Public Properties

Name	Description
 ActiveSketch	Gets the active sketch.
 AddToDB	Gets or sets whether sketch entities are added directly to the SOLIDWORKS database.
 AutoInference	Obsolete. Superseded by ISldWorks::GetUserPreferenceToggle or ISldWorks::SetUserPreferenceToggle and swUserPreferenceToggle_e.swSketchInference .
 AutoSolve	Gets or sets whether SOLIDWORKS automatically solves the sketch geometry of the part while creating it.
 CurvatureDensity	Gets or sets the scaling factor by which to adjust the density of the curvature combs for this spline.
 CurvatureScale	Gets or sets the scaling factor by which to adjust the size of the curvature combs for this spline.
 DisplayWhenAdded	Gets or sets whether new sketch entities are immediately displayed when created.
 Document	Gets the document for this ISketchManager object.
 InferenceMode	Obsolete. Superseded by ISldWorks::GetUserPreferenceToggle or ISldWorks::SetUserPreferenceToggle and swUserPreferenceToggle_e.swSketchInference .

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See Also

[ISketchManager Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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ActiveSketch Property (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : ActiveSketch Property (ISketchManager)

Gets the active sketch.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ActiveSketch As Sketch
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As Sketch  
  
value = instance.ActiveSketch
```

C#

```
Sketch ActiveSketch {get;}
```

C++/CLI

```
property Sketch^ ActiveSketch {  
    Sketch^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Active [sketch](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager.ActiveSketch](#).

Example

[Constrain Sketch \(VBA\)](#)
[Create 3D Sketch Plane \(C#\)](#)
[Create 3D Sketch Plane \(VB.NET\)](#)
[Create 3D Sketch Plane \(VBA\)](#)
[Rotate and Copy 3D Sketch About Coordinates \(C#\)](#)
[Rotate and Copy 3D Sketch About Coordinates \(VB.NET\)](#)
[Rotate and Copy 3D Sketch About Coordinates \(VBA\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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AddAlongXDimension Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : AddAlongXDimension Method (ISketchManager)

Projects and displays along the x axis a dimension between selected points in a 3D sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function AddAlongXDimension( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As DisplayDimension  
  
value = instance.AddAlongXDimension(X, Y, Z)
```

C#

```
DisplayDimension AddAlongXDimension(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
DisplayDimension^ AddAlongXDimension(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X coordinate of dimension text placement
Y
Y coordinate of dimension text placement
Z
Z coordinate of dimension text placement

Return Value

[IDisplayDimension](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::AddAlongXDimension](#).

Example

[Add Along X Dimension to 3D Sketch \(VBA\)](#)
[Add Along X Dimension to 3D Sketch \(VB.NET\)](#)
[Add Along X Dimension to 3D Sketch \(C#\)](#)

Remarks

This method works only for 3D sketches in edit mode, and the display dimension is visible only when the sketch is in edit mode.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::FullyDefineSketch Method](#)
[ISketchManager::AddAlongYDimension Method](#)
[ISketchManager::AddAlongZDimension Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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AddAlongYDimension Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : AddAlongYDimension Method (ISketchManager)

Projects and displays along the y axis a dimension between selected points in a 3D sketch.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function AddAlongYDimension( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As DisplayDimension  
  
value = instance.AddAlongYDimension(X, Y, Z)
```

C#

```
DisplayDimension AddAlongYDimension(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
DisplayDimension^ AddAlongYDimension(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X coordinate of dimension text placement
Y
Y coordinate of dimension text placement
Z
Z coordinate of dimension text placement

Return Value

[IDisplayDimension](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::AddAlongYDimension](#).

Example

[Add Along Y Dimension to 3D Sketch \(VBA\)](#)
[Add Along Y Dimension to 3D Sketch \(VB.NET\)](#)
[Add Along Y Dimension to 3D Sketch \(C#\)](#)

Remarks

This method works only for 3D sketches in edit mode, and the display dimension is visible only when the sketch is in edit mode.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::FullyDefineSketch Method](#)
[ISketchManager::AddAlongXDimension Method](#)
[ISketchManager::AddAlongZDimension Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

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AddAlongZDimension Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : AddAlongZDimension Method (ISketchManager)

Projects and displays along the z axis a dimension between selected points in a 3D sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function AddAlongZDimension( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As DisplayDimension
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As DisplayDimension
```

```
value = instance.AddAlongZDimension(X, Y, Z)
```

C#

```
DisplayDimension AddAlongZDimension(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
DisplayDimension^ AddAlongZDimension(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X](#)
X coordinate of dimension text placement
[Y](#)
Y coordinate of dimension text placement
[Z](#)
Z coordinate of dimension text placement

Return Value

Visual Basic for Applications (VBA) Syntax

See [SketchManager::AddAlongZDimension](#).

Example

[Add Along Z Dimension to 3D Sketch \(VBA\)](#)
[Add Along Z Dimension to 3D Sketch \(VB.NET\)](#)
[Add Along Z Dimension to 3D Sketch \(C#\)](#)

Remarks

This method works only for 3D sketches in edit mode, and the display dimension is visible only when the sketch is in edit mode.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::AddAlongXDimension Method](#)
[ISketchManager::AddAlongYDimension Method](#)
[ISketchManager::FullyDefineSketch Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddToDB Property (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : AddToDB Property (ISketchManager)

Gets or sets whether sketch entities are added directly to the SOLIDWORKS database.

.NET Syntax

Visual Basic (Declaration)

Property AddToDB As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As System.Boolean
```

```
instance.AddToDB = value
```

```
value = instance.AddToDB
```

C#

```
System.bool AddToDB {get; set;}
```

C++/CLI

```
property System.bool AddToDB {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to add items directly to the SOLIDWORKS database, false to not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::AddToDb](#).

Example

[Get Assembly Bounding Box Using Components \(C#\)](#)
[Get Assembly Bounding Box Using Components \(VB.NET\)](#)
[Get Assembly Bounding Box Using Components \(VBA\)](#)

Remarks

One of the benefits of adding sketch entities directly to the database is that you can avoid grid and entity snapping. For example, if you create a sketch line whose endpoint is near another entity or grid point, the new sketch line endpoint snaps to the other entity or grid point. Setting this property to true avoids this behavior during sketch entity creation.

After setting this property to true:

- It is not possible to undo the creation of some types of sketch entities.
- You must reset it to false to restore SOLIDWORKS' normal operating mode.

This property and [ISketchManager::DisplayWhenAdded](#) improve performance during sketch entity creation. [ISketchManager::DisplayWhenAdded](#) requires that this property be set to true.

If you want to constrain all the sketch entities after creation, use [ISketch::ConstrainAll](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

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AutoInference Property (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : AutoInference Property (ISketchManager)

Obsolete. Superseded by [ISldWorks::GetUserPreferenceToggle](#) or [ISldWorks::SetUserPreferenceToggle](#) and [swUserPreferenceToggle.e_swSketchInference](#).

.NET Syntax

Visual Basic (Declaration)

Property AutoInference As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As System.Boolean
```

```
instance.AutoInference = value
```

```
value = instance.AutoInference
```

C#

```
System.bool AutoInference {get; set;}
```

C++/CLI

```
property System.bool AutoInference {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if automatic inferencing is on, false if off

Visual Basic for Applications (VBA) Syntax

See [SketchManager::AutoInference](#).

Remarks

Inferencing mode can be seen when creating a sketch segment and your cursor moves past another sketch entity. If inferencing is turned on, you see a dashed line from your current cursor position to the inferred position on the existing sketch entity.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::InferenceMode Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AutoSolve Property (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : AutoSolve Property (ISketchManager)

Gets or sets whether SOLIDWORKS automatically solves the sketch geometry of the part while creating it.

.NET Syntax

Visual Basic (Declaration)

Property AutoSolve As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim value As System.Boolean
```

```
instance.AutoSolve = value
```

```
value = instance.AutoSolve
```

C#

```
System.bool AutoSolve {get; set;}
```

C++/CLI

```
property System.bool AutoSolve {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if automatic solve is on, false if off

Visual Basic for Applications (VBA) Syntax

See [SketchManager::AutoSolve](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketch::GetAutomaticSolve Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ConvertEntities Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : ConvertEntities Method (ISketchManager)

Not implemented. Use [ISketchManager::SketchUseEdge2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ConvertEntities()
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
  
instance.ConvertEntities()
```

C#

```
void ConvertEntities()
```

C++/CLI

```
void ConvertEntities();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Create3PointArc Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : Create3PointArc Method (ISketchManager)

Creates a 3-point arc.

.NET Syntax

Visual Basic (Declaration)

```
Function Create3PointArc( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double, _  
    ByVal X3 As System.Double, _  
    ByVal Y3 As System.Double, _  
    ByVal Z3 As System.Double _  
) As SketchSegment
```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim X1 As System.Double
Dim Y1 As System.Double
Dim Z1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim Z2 As System.Double
Dim X3 As System.Double
Dim Y3 As System.Double
Dim Z3 As System.Double
Dim value As SketchSegment

value = instance.Create3PointArc(X1, Y1, Z1, X2, Y2, Z2, X3, Y3, Z3)

```

C#

```

SketchSegment Create3PointArc(
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2,
    System.double X3,
    System.double Y3,
    System.double Z3
)

```

C++/CLI

```

SketchSegment^ Create3PointArc(
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2,
    & System.double X3,
    & System.double Y3,
    & System.double Z3
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X1
 X coordinate of point 1
Y1
 Y coordinate of point 1
Z1
 Z coordinate of point 1
X2
 X coordinate of point 2
Y2
 Y coordinate of point 2
Z2
 Z coordinate of point 2
X3
 X coordinate of point 3
Y3
 Y coordinate of point 3
Z3
 Z coordinate of point 3

Return Value

[Sketch segment](#) for the 3-point arc

Example

[Create 3-Point Arc \(VBA\)](#)
[Create 3-Point Arc \(VB.NET\)](#)
[Create 3-Point Arc \(C#\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateArc Method](#)
[ISketchManager::CreateTangentArc Method](#)
[IModelDoc2::CreateArcByCenter Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Create3PointCenterRectangle Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : Create3PointCenterRectangle Method (ISketchManager)

Creates a 3-point center rectangle at any angle.

.NET Syntax

Visual Basic (Declaration)

```
Function Create3PointCenterRectangle( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double, _  
    ByVal X3 As System.Double, _  
    ByVal Y3 As System.Double, _  
    ByVal Z3 As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
Dim X3 As System.Double  
Dim Y3 As System.Double  
Dim Z3 As System.Double  
Dim value As System.Object
```

```
value = instance.Create3PointCenterRectangle(X1, Y1, Z1, X2, Y2, Z2, X3, Y3, Z3)
```

C#

```
System.object Create3PointCenterRectangle(  
    System.double X1,  
    System.double Y1,  
    System.double Z1,  
    System.double X2,  
    System.double Y2,  
    System.double Z2,  
    System.double X3,  
    System.double Y3,
```

```
        System.double Z3  
    )
```

C++/CLI

```
System.Object^ Create3PointCenterRectangle(  
& System.double X1,  
& System.double Y1,  
& System.double Z1,  
& System.double X2,  
& System.double Y2,  
& System.double Z2,  
& System.double X3,  
& System.double Y3,  
& System.double Z3  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X1](#)
X coordinate of point 1
[Y1](#)
Y coordinate of point 1

[Z1](#)
Z coordinate of point 1
[X2](#)
X coordinate of point 2
[Y2](#)
Y coordinate of point 2
[Z2](#)
Z coordinate of point 2
[X3](#)
X coordinate of point 3
[Y3](#)
Y coordinate of point 3
[Z3](#)
Z coordinate of point 3

Return Value

Array of [sketch segments](#) that represent the edges and diagonals created for this center rectangle

Visual Basic for Applications (VBA) Syntax

See [SketchManager::Create3PointCenterRectangle](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::Create3PointCornerRectangle Method](#)
[ISketchManager::CreateCenterRectangle Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Create3PointCornerRectangle Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : Create3PointCornerRectangle Method (ISketchManager)

Creates a 3-point corner rectangle at any angle.

.NET Syntax

Visual Basic (Declaration)

```
Function Create3PointCornerRectangle( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _
```

```

        ByVal Y2 As System.Double, _
        ByVal Z2 As System.Double, _
        ByVal X3 As System.Double, _
        ByVal Y3 As System.Double, _
        ByVal Z3 As System.Double _
    ) As System.Object

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim X1 As System.Double
Dim Y1 As System.Double
Dim Z1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim Z2 As System.Double
Dim X3 As System.Double
Dim Y3 As System.Double
Dim Z3 As System.Double
Dim value As System.Object

```

```
value = instance.Create3PointCornerRectangle(X1, Y1, Z1, X2, Y2, Z2, X3, Y3, Z3)
```

C#

```

System.object Create3PointCornerRectangle(
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2,
    System.double X3,
    System.double Y3,
    System.double Z3
)

```

C++/CLI

```

System.Object^ Create3PointCornerRectangle(
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2,
    & System.double X3,
    & System.double Y3,
    & System.double Z3
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X1
 X coordinate of point 1
Y1
 Y coordinate of point 1
Z1
 Z coordinate of point 1
X2
 X coordinate of point 2
Y2
 Y coordinate of point 2
Z2
 Z coordinate of point 2
X3
 X coordinate of point 3
Y3
 Y coordinate of point 3
Z3
 Z coordinate of point 3

Return Value

Array of [sketch segments](#) that represent the edges created for this corner rectangle

Visual Basic for Applications (VBA) Syntax

See [SketchManager::Create3PointCornerRectangle](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::Create3PointCenterRectangle Method](#)
[ISketchManager::CreateCenterRectangle Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateArc Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateArc Method (ISketchManager)

Creates an arc based on a center point, a start point, an end point, and a direction.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateArc( _  
    ByVal XC As System.Double, _  
    ByVal YC As System.Double, _  
    ByVal Zc As System.Double, _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double, _  
    ByVal Direction As System.Short _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim XC As System.Double  
Dim YC As System.Double  
Dim Zc As System.Double  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
Dim Direction As System.Short  
Dim value As SketchSegment
```

```
value = instance.CreateArc(XC, YC, Zc, X1, Y1, Z1, X2, Y2, Z2, Direction)
```

C#

```
SketchSegment CreateArc(  
    System.double XC,  
    System.double YC,  
    System.double Zc,  
    System.double X1,  
    System.double Y1,  
    System.double Z1,  
    System.double X2,  
    System.double Y2,
```

```
        System.double Z2,  
        System.short Direction  
    )
```

C++/CLI

```
SketchSegment^ CreateArc(  
& System.double XC,  
& System.double YC,  
& System.double Zc,  
& System.double X1,  
& System.double Y1,  
& System.double Z1,  
& System.double X2,  
& System.double Y2,  
& System.double Z2,  
& System.short Direction  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XC

X coordinate of the circle center point in meters

YC

coordinate of the circle center point in meters

Zc

Z coordinate of the circle center point in meters

X1

X coordinate of the start point of the arc in meters

Y1

coordinate of the start point of the arc in meters

Z1

Z coordinate of the start point of the arc in meters

X2

X coordinate of the end point of the arc in meters

Y2

Y coordinate of the end point of the arc in meters

Z2

coordinate of the end point of the arc in meters

Direction

- +1 : Go from the start point to the end point in a counter-clockwise direction
- -1 : Go from the start point to the end point in a clockwise direction

Return Value

[ISketchArc](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateArc](#).

Example

[Get Sketch Arc Data \(VBA\)](#)

[Get Sketch Arc Data \(VB.NET\)](#)

[Get Sketch Arc Data \(C#\)](#)

Remarks

This method creates a partial arc in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [ISketchManager::ActiveSketch](#).

[ISketchManager::AddToDB](#) and [ISketchManager::DisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

[ISketchManager::AddToDB](#) also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also increases the performance of this API. When you are done creating entities, it is important to call [ISketchManager::AddToDB\(False\)](#), to restore SOLIDWORKS to its normal operating mode.

This method also works with [ISketchManager::DisplayWhenAdded](#). If you have called [ISketchManager::AddToDB \(True\)](#), additional performance can be gained by calling [ISketchManager::DisplayWhenAdded\(False\)](#) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelView::GraphicsRedraw](#) or [IModelView::IGraphicsRedraw](#)) to see the entities that you added. You should also restore the original display settings by calling [ISketchManager::DisplayWhenAdded\(True\)](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[IModelDoc2::CreateArcByCenter Method](#)
[ISketchManager::Create3PointArc Method](#)
[ISketchManager::CreateTangentArc Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateBoundaryHatch Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateBoundaryHatch Method (ISketchManager)

Creates area hatch/fill boundary hatches using closed sketch profiles.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBoundaryHatch( _  
    ByVal Angle As System.Double, _  
    ByVal Scale As System.Double, _  
    ByVal Color As System.Integer, _  
    ByVal Hatchname As System.String, _  
    ByVal Layername As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Angle As System.Double  
Dim Scale As System.Double  
Dim Color As System.Integer  
Dim Hatchname As System.String  
Dim Layername As System.String  
Dim value As System.Object  
  
value = instance.CreateBoundaryHatch(Angle, Scale, Color, Hatchname, Layername)
```

C#

```
System.object CreateBoundaryHatch(  
    System.double Angle,  
    System.double Scale,  
    System.int Color,  
    System.string Hatchname,  
    System.string Layername  
)
```

C++/CLI

```
System.Object^ CreateBoundaryHatch(  
& System.double Angle,  
& System.double Scale,  
& System.int Color,  
& System.String^ Hatchname,  
& System.String^ Layername  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

Angle of the hatch if hatch is not solid

Scale

Scale factor for the hatch if hatch is not solid

Color

COLORREF value describing the color used for this hatch

Hatchname

Type or name of the hatch from the **Sldwks.ptn** file

Layername

Layer name for the hatch if a drawing document

Return Value

Array of [sketch hatches](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateBoundaryHatch](#).

Remarks

Before calling this method, select the closed sketch profile for the area hatch/fill boundary hatch.

See Also

[ISketchManager Interface](#)

[ISketchManager Members](#)

[ISketchManager::CreateRegionHatch Method](#)

[IModelDoc2::InsertHatchedFace Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCenterLine Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateCenterLine Method (ISketchManager)

Creates a center line between the specified points.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCenterLine( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
Dim value As SketchSegment
```

```
value = instance.CreateCenterLine(X1, Y1, Z1, X2, Y2, Z2)
```

C#

```
SketchSegment CreateCenterLine(  
    System.double X1,  
    System.double Y1,  
    System.double Z1,  
    System.double X2,
```

```
System.double Y2,  
System.double Z2  
)
```

C++/CLI

```
SketchSegment^ CreateCenterLine(  
& System.double X1,  
& System.double Y1,  
& System.double Z1,  
& System.double X2,  
& System.double Y2,  
& System.double Z2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X1
X coordinate of first end point, in meters
Y1
Y coordinate of first end point, in meters
Z1
Z coordinate of first end point, in meters
X2
X coordinate of second end point, in meters
Y2
Y coordinate of second end point, in meters
Z2
Z coordinate of second end point, in meters

Return Value

[Sketch segment](#) for the center line

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateCenterLine](#).

Example

[Create 3D Sketch Plane \(C#\)](#)
[Create 3D Sketch Plane \(VB.NET\)](#)
[Create 3D Sketch Plane \(VBA\)](#)

Remarks

You can also create centerline construction geometry using [ISketchManager::CreateLine](#) and [ISketchSegment::ConstructionGeometry](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCenterRectangle Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateCenterRectangle Method (ISketchManager)

Creates a center rectangle.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCenterRectangle( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _
```

```

        ByVal Z1 As System.Double, _
        ByVal X2 As System.Double, _
        ByVal Y2 As System.Double, _
        ByVal Z2 As System.Double _
    ) As System.Object

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim X1 As System.Double
Dim Y1 As System.Double
Dim Z1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim Z2 As System.Double
Dim value As System.Object

value = instance.CreateCenterRectangle(X1, Y1, Z1, X2, Y2, Z2)

```

C#

```

System.object CreateCenterRectangle(
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2
)

```

C++/CLI

```

System.Object^ CreateCenterRectangle(
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X1
 X coordinate for point 1
Y1
 Y coordinate for point 1
Z1
 Z coordinate for point 1
X2
 X coordinate for point 2
Y2
 Y coordinate for point 2
Z2
 Z coordinate for point 2

Return Value

Array of [sketch segments](#) that represent the edges and diagonals created for this center rectangle

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateCenterRectangle](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::Create3PointCenterRectangle Method](#)
[ISketchManager::Create3PointCornerRectangle Method](#)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateChamfer Method (ISketchManager)

Creates a chamfer between two selected sketch entities.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateChamfer( _  
    ByVal Type As System.Integer, _  
    ByVal Distance As System.Double, _  
    ByVal AngleORdist As System.Double _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Type As System.Integer  
Dim Distance As System.Double  
Dim AngleORdist As System.Double  
Dim value As SketchSegment  
  
value = instance.CreateChamfer(Type, Distance, AngleORdist)
```

C#

```
SketchSegment CreateChamfer(  
    System.int Type,  
    System.double Distance,  
    System.double AngleORdist  
)
```

C++/CLI

```
SketchSegment^ CreateChamfer(  
& System.int Type,  
& System.double Distance,  
& System.double AngleORdist  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type of chamfer as defined in [swSketchChamferType e](#)

Distance

Distance of the chamfer

AngleORdist

- If Type = swSketchChamfer_DistanceDistance, then the second chamfer distance
- If Type = swSketchChamfer_DistanceAngle, then the second chamfer angle
- If Type = swSketchChamfer_DistanceEqual, then this argument is ignored because Distance is used for both edges

Return Value

[Sketch segment](#) for the chamfer

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateChamfer](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

CreateCircle Method (ISketchManager)

[See Also](#) [Example](#)[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateCircle Method (ISketchManager)

Creates a circle based on a center point and a point on the circle.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCircle( _
    ByVal XC As System.Double, _
    ByVal YC As System.Double, _
    ByVal Zc As System.Double, _
    ByVal Xp As System.Double, _
    ByVal Yp As System.Double, _
    ByVal Zp As System.Double _
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim XC As System.Double
Dim YC As System.Double
Dim Zc As System.Double
Dim Xp As System.Double
Dim Yp As System.Double
Dim Zp As System.Double
Dim value As SketchSegment
```

```
value = instance.CreateCircle(XC, YC, Zc, Xp, Yp, Zp)
```

C#

```
SketchSegment CreateCircle(
    System.double XC,
    System.double YC,
    System.double Zc,
    System.double Xp,
    System.double Yp,
    System.double Zp
)
```

C++/CLI

```
SketchSegment^ CreateCircle(
    & System.double XC,
    & System.double YC,
    & System.double Zc,
    & System.double Xp,
    & System.double Yp,
    & System.double Zp
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[XC](#)
X coordinate of the circle center point, in meters

[YC](#)
Y coordinate of the circle center point, in meters

[Zc](#)
Z coordinate of the circle center point, in meters

[Xp](#)
X coordinate of the point on the circle, in meters

[Yp](#)
Y coordinate of the point on the circle, in meters

[Zp](#)
Z coordinate of the point on the circle, in meters

Return Value

[Sketch segment](#) for the circle

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateCircle](#).

Example

[Create and Edit Circular Sketch Pattern \(VB.NET\)](#)
[Create and Edit Circular Sketch Pattern \(VBA\)](#)
[Create and Edit Circular Sketch Pattern \(C#\)](#)
[Create Detail Circle and Detail View \(C#\)](#)
[Create Detail Circle and Detail View \(VB.NET\)](#)
[Create Detail Circle and Detail View \(VBA\)](#)

Remarks

This method creates a full circle in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [ISketchManager::ActiveSketch](#).

[ISketchManager::AddToDB](#) and [ISketchManager::DisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

ISketchManager::AddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also significantly increases the performance of this method. When you are done creating entities, it is important to [ISketchManager::AddToDB\(False\)](#), to restore SOLIDWORKS to its normal operating mode.

This method also works with [ISketchManager::DisplayWhenAdded](#). If you have called [ISketchManager::AddToDB\(True\)](#), additional performance can be gained by calling [ISketchManager::DisplayWhenAdded\(False\)](#) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelView::GraphicsRedraw](#) or [IModelView::IGraphicsRedraw](#)) to see the entities you added. You should also restore the original display settings by calling [ISketchManager::DisplayWhenAdded\(True\)](#).

To create a circle using a center point and radius, see [ISketchManager::CreateCircleByRadius](#). To create a partial arc, see [ISketchManager::CreateArc](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateConic Method \(\)](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCircleByRadius Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateCircleByRadius Method (ISketchManager)

Creates a circle based on a center point and a specified radius.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCircleByRadius( _  
    ByVal XC As System.Double, _  
    ByVal YC As System.Double, _  
    ByVal Zc As System.Double, _  
    ByVal Radius As System.Double _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim XC As System.Double  
Dim YC As System.Double  
Dim Zc As System.Double
```

```
Dim Radius As System.Double
Dim value As SketchSegment
```

```
value = instance.CreateCircleByRadius(XC, YC, Zc, Radius)
```

C#

```
SketchSegment CreateCircleByRadius(
    System.double XC,
    System.double YC,
    System.double Zc,
    System.double Radius
)
```

C++/CLI

```
SketchSegment^ CreateCircleByRadius(
    & System.double XC,
    & System.double YC,
    & System.double Zc,
    & System.double Radius
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XC
X coordinate of the circle center point in meters
YC
Y coordinate of the circle center point in meters
Zc
Z coordinate of the circle center point in meters
Radius
Radius of the circle in meters

Return Value

[Sketch segment](#) for the circle

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateCircle ByRadius](#).

Remarks

This method creates a full circle in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [ISketchManager::ActiveSketch](#).

[ISketchManager::AddToDB](#) and [ISketchManager::DisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

ISketchManager::AddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also significantly increases the performance of this method. When you are done creating entities, it is important to ISketchManager::AddToDB(False), to restore SOLIDWORKS to its normal operating mode.

This method also works with ISketchManager::DisplayWhenAdded. If you have called ISketchManager::AddToDB(True), additional performance can be gained by calling ISketchManager::DisplayWhenAdded(False) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelView::GraphicsRedraw](#) or [IModelView::IGraphicsRedraw](#)) to see the entities you added. You should also restore the original display settings by calling ISketchManager::DisplayWhenAdded(True).

To create a circle using a center point and a point on the circle, see [ISketchManager::CreateCircle](#). To create a partial arc, see [ISketchManager::CreateArc](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

CreateCircularSketchStepAndRepeat Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateCircularSketchStepAndRepeat Method (ISketchManager)

Creates circular sketch pattern.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCircularSketchStepAndRepeat( _  
    ByVal ArcRadius As System.Double, _  
    ByVal ArcAngle As System.Double, _  
    ByVal PatternNum As System.Integer, _  
    ByVal PatternSpacing As System.Double, _  
    ByVal PatternRotate As System.Boolean, _  
    ByVal DeleteInstances As System.String, _  
    ByVal RadiusDim As System.Boolean, _  
    ByVal AngleDim As System.Boolean, _  
    ByVal CreateNumOfInstancesDim As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim ArcRadius As System.Double  
Dim ArcAngle As System.Double  
Dim PatternNum As System.Integer  
Dim PatternSpacing As System.Double  
Dim PatternRotate As System.Boolean  
Dim DeleteInstances As System.String  
Dim RadiusDim As System.Boolean  
Dim AngleDim As System.Boolean  
Dim CreateNumOfInstancesDim As System.Boolean  
Dim value As System.Boolean  
  
value = instance.CreateCircularSketchStepAndRepeat(ArcRadius, ArcAngle, PatternNum,  
PatternSpacing, PatternRotate, DeleteInstances, RadiusDim, AngleDim, CreateNumOfInstancesDim)
```

C#

```
System.bool CreateCircularSketchStepAndRepeat(  
    System.double ArcRadius,  
    System.double ArcAngle,  
    System.int PatternNum,  
    System.double PatternSpacing,  
    System.bool PatternRotate,  
    System.string DeleteInstances,  
    System.bool RadiusDim,  
    System.bool AngleDim,  
    System.bool CreateNumOfInstancesDim  
)
```

C++/CLI

```
System.bool CreateCircularSketchStepAndRepeat(  
& System.double ArcRadius,  
& System.double ArcAngle,  
& System.int PatternNum,  
& System.double PatternSpacing,  
& System.bool PatternRotate,  
& System.String^ DeleteInstances,  
& System.bool RadiusDim,  
& System.bool AngleDim,  
& System.bool CreateNumOfInstancesDim  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ArcRadius

Radius for the circular sketch pattern

ArcAngle

Angle relative to the sketch entities being patterned

PatternNum

Total number of instances, including the seed geometry

PatternSpacing

Spacing between pattern instances

PatternRotate

True to rotate the pattern, false to not

DeleteInstances

Number of instances to delete, passed as a string formatted as: "(a) (b) (c) "

RadiusDim

True to display the radius dimension in the graphics area, false to not

AngleDim

True to display the angle dimension in the graphics area, false to not

CreateNumOfInstancesDim

True to display the number of instances dimension in the graphics area, false to not

Return Value

True if the circular sketch is created, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateCircularSketchStepAndRepeat](#).

Example

[Create and Edit Circular Sketch Pattern \(VB.NET\)](#)

[Create and Edit Circular Sketch Pattern \(VBA\)](#)

[Create and Edit Circular Sketch Pattern \(C#\)](#)

Remarks

The radius, angle, and number of instances dimensions displayed in the graphics area can be modified by interactive users.

See Also

[ISketchManager Interface](#)

[ISketchManager Members](#)

[ISketchManager::EditCircularSketchStepAndRepeat Method](#)

[ISketchManager::CreateLinearSketchStepAndRepeat Method](#)

[ISketchManager::EditLinearSketchStepAndRepeat Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateConic Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateConic Method (ISketchManager)

Creates a conic curve in the active sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateConic( _  
    ByVal XFocus As System.Double, _  
    ByVal YFocus As System.Double, _  
    ByVal ZFocus As System.Double, _  
    ByVal XApex As System.Double, _  
    ByVal YApex As System.Double, _  
    ByVal ZApex As System.Double, _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _
```

```

        ByVal Y2 As System.Double, _
        ByVal Z2 As System.Double _
    ) As SketchSegment

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim XFocus As System.Double
Dim YFocus As System.Double
Dim ZFocus As System.Double
Dim XApex As System.Double
Dim YApex As System.Double
Dim ZApex As System.Double
Dim X1 As System.Double
Dim Y1 As System.Double
Dim Z1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim Z2 As System.Double
Dim value As SketchSegment

```

```

value = instance.CreateConic(XFocus, YFocus, ZFocus, XApex, YApex, ZApex, X1, Y1, Z1, X2, Y2, Z2)

```

C#

```

SketchSegment CreateConic(
    System.double XFocus,
    System.double YFocus,
    System.double ZFocus,
    System.double XApex,
    System.double YApex,
    System.double ZApex,
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2
)

```

C++/CLI

```

SketchSegment^ CreateConic(
    & System.double XFocus,
    & System.double YFocus,
    & System.double ZFocus,
    & System.double XApex,
    & System.double YApex,
    & System.double ZApex,
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XFocus
X coordinate for the focus of the curve

YFocus
Y coordinate for the focus of the curve

ZFocus
Z coordinate for the focus of the curve

XApex
X coordinate for the apex of the curve

YApex
Y coordinate for the apex of the curve

ZApex
Z coordinate for the apex of the curve

X1
X coordinate for the start point of the curve

Y1

Y coordinate for the start point of the curve
Z1
Z coordinate for the start point of the curve
X2
X coordinate for the end point of the curve
Y2
Y coordinate for the end point of the curve
Z2
Z coordinate for the end point of the curve

Return Value

[Sketch segment](#) for the conic curve

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateConic](#).

Example

[Insert Conic Curve Example \(VBA\)](#)
[Insert Conic Curve \(VB.NET\)](#)
[Insert Conic Curve \(C#\)](#)

Remarks

This method creates a conic curve (circle, ellipse, parabola, or hyperbola) in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [ISketchManager::ActiveSketch](#).

[ISketchManager::AddToDB](#) and [ISketchManager::DisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

ISketchManager::AddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also significantly increases the performance of this method. When you are done creating entities, it is important to call ISketchManager::AddToDB(False), to restore SOLIDWORKS to its normal operating mode.

This method also works with ISketchManager::DisplayWhenAdded. If you have called ISketchManager::AddToDB(True), additional performance can be gained by calling ISketchManager::DisplayWhenAdded(False) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelView::GraphicsRedraw](#) or [IModelView::IGraphicsRedraw](#)) to see the entities you added. You should also restore the original display settings by calling ISketchManager::DisplayWhenAdded(True).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateCircle Method \(\)](#)
[ISketchManager::CreateEllipse Method \(\)](#)
[ISketchManager::CreateParabola Method \(\)](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateConstructionGeometry Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateConstructionGeometry Method (ISketchManager)

Sets selected sketch segments to be construction geometry instead of sketch geometry.

.NET Syntax

Visual Basic (Declaration)

Sub CreateConstructionGeometry()

Visual Basic (Usage)



```
Dim instance As ISketchManager  
instance.CreateConstructionGeometry()
```

C#

```
void CreateConstructionGeometry()
```

C++/CLI

```
void CreateConstructionGeometry();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateConstructionGeometry](#).

Remarks

This method supports all document types, unlike [IDrawingDoc::CreateConstructionGeometry](#), which only supports drawing documents. Both methods perform the same functionality.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCornerRectangle Method (ISketchManager)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateCornerRectangle Method (ISketchManager)

Creates a corner rectangle.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCornerRectangle( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double _  
    ) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
Dim value As System.Object  
  
value = instance.CreateCornerRectangle(X1, Y1, Z1, X2, Y2, Z2)
```

C#

```
System.Object CreateCornerRectangle(
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2
)
```

C++/CLI

```
System.Object^ CreateCornerRectangle(
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X1](#)
Upper-left X coordinate for point 1
[Y1](#)
Upper-left Y coordinate for point 1
[Z1](#)
Upper-left Z coordinate for point 1
[X2](#)
Lower-right X coordinate for point 2
[Y2](#)
Lower-right Y coordinate for point 2
[Z2](#)
Lower-right Z coordinate for point 2

Return Value

Array of [sketch segments](#) that represent the edges created for this corner rectangle

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateCornerRectangle](#).

Example

[Create and Edit Circular Sketch Pattern \(VB.NET\)](#)
[Create and Edit Circular Sketch Pattern \(VBA\)](#)
[Create and Edit Circular Sketch Pattern \(C#\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::Create3PointCenterRectangle Method](#)
[ISketchManager::Create3PointCornerRectangle Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateEllipse Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateEllipse Method (ISketchManager)

Creates an ellipse using the specified center, major-axis, and minor-axis points.

.NET Syntax

Visual Basic (Declaration)

```

Function CreateEllipse( _
    ByVal XC As System.Double, _
    ByVal YC As System.Double, _
    ByVal Zc As System.Double, _
    ByVal XMajor As System.Double, _
    ByVal YMajor As System.Double, _
    ByVal ZMajor As System.Double, _
    ByVal XMinor As System.Double, _
    ByVal YMinor As System.Double, _
    ByVal ZMinor As System.Double _
) As SketchSegment

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim XC As System.Double
Dim YC As System.Double
Dim Zc As System.Double
Dim XMajor As System.Double
Dim YMajor As System.Double
Dim ZMajor As System.Double
Dim XMinor As System.Double
Dim YMinor As System.Double
Dim ZMinor As System.Double
Dim value As SketchSegment

```

```
value = instance.CreateEllipse(XC, YC, Zc, XMajor, YMajor, ZMajor, XMinor, YMinor, ZMinor)
```

C#

```

SketchSegment CreateEllipse(
    System.double XC,
    System.double YC,
    System.double Zc,
    System.double XMajor,
    System.double YMajor,
    System.double ZMajor,
    System.double XMinor,
    System.double YMinor,
    System.double ZMinor
)

```

C++/CLI

```

SketchSegment^ CreateEllipse(
    & System.double XC,
    & System.double YC,
    & System.double Zc,
    & System.double XMajor,
    & System.double YMajor,
    & System.double ZMajor,
    & System.double XMinor,
    & System.double YMinor,
    & System.double ZMinor
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[XC](#)
 X coordinate of the ellipse center point
[YC](#)
 Y coordinate of the ellipse center point
[Zc](#)
 Z coordinate of the ellipse center point
[XMajor](#)
 X coordinate of the point on the ellipse that is on the major axis
[YMajor](#)
 Y coordinate of the point on the ellipse that is on the major axis
[ZMajor](#)
 Z coordinate of the point on the ellipse that is on the major axis
[XMinor](#)
 X coordinate of the point on the ellipse that is on the minor axis
[YMinor](#)

Y coordinate of the point on the ellipse that is on the minor axis
ZMinor
Z coordinate of the point on the ellipse that is on the minor axis

Return Value

[Sketch segment](#) for the ellipse

 Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateEllipse](#).

 Example

[Create Ellipse \(VBA\)](#)
[Create Ellipse \(VB.NET\)](#)
[Create Ellipse \(C#\)](#)

 Remarks

This method creates a full ellipse in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [ISketchManager::ActiveSketch](#).

[ISketchManager::AddToDB](#) and [ISketchManager::DisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

ISketchManager::AddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also significantly increases the performance of this method. When you are done creating entities, it is important to [ISketchManager::AddToDB\(False\)](#), to restore SOLIDWORKS to its normal operating mode.

This method also works with [ISketchManager::DisplayWhenAdded](#). If you have called [ISketchManager::AddToDB\(True\)](#), additional performance can be gained by calling [ISketchManager::DisplayWhenAdded\(False\)](#) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelView::GraphicsRedraw](#) or [IModelView::IGraphicsRedraw](#)) to see the entities you added. You should also restore the original display settings by calling [ISketchManager::DisplayWhenAdded\(True\)](#).

To create a partial ellipse, use [ISketchManager::CreateEllipticalArc](#).

 See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateConic Method \(\)](#)

 Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateEllipticalArc Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateEllipticalArc Method (ISketchManager)

Creates a partial ellipse given a center point, two points that specify the major and minor axis, and two points that define the elliptical start and end points.

 .NET Syntax

Visual Basic (Declaration)

```
Function CreateEllipticalArc( _  
    ByVal XC As System.Double, _  
    ByVal YC As System.Double, _  
    ByVal Zc As System.Double, _  
    ByVal XMajor As System.Double, _  
    ByVal YMajor As System.Double, _  
    ByVal ZMajor As System.Double, _  
    ByVal XMinor As System.Double, _  
    ByVal YMinor As System.Double, _  
    ByVal ZMinor As System.Double, _  
    ByVal XI As System.Double, _  
    ByVal YI As System.Double, _
```

```

        ByVal Z1 As System.Double, _
        ByVal X2 As System.Double, _
        ByVal Y2 As System.Double, _
        ByVal Z2 As System.Double, _
        ByVal Direction As System.Short _
    ) As SketchSegment

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim XC As System.Double
Dim YC As System.Double
Dim Zc As System.Double
Dim XMajor As System.Double
Dim YMajor As System.Double
Dim ZMajor As System.Double
Dim XMinor As System.Double
Dim YMinor As System.Double
Dim ZMinor As System.Double
Dim X1 As System.Double
Dim Y1 As System.Double
Dim Z1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim Z2 As System.Double
Dim Direction As System.Short
Dim value As SketchSegment

```

```

value = instance.CreateEllipticalArc(XC, YC, Zc, XMajor, YMajor, ZMajor, XMinor, YMinor, ZMinor,
X1, Y1, Z1, X2, Y2, Z2, Direction)

```

C#

```

SketchSegment CreateEllipticalArc(
    System.double XC,
    System.double YC,
    System.double Zc,
    System.double XMajor,
    System.double YMajor,
    System.double ZMajor,
    System.double XMinor,
    System.double YMinor,
    System.double ZMinor,
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2,
    System.short Direction
)

```

C++/CLI

```

SketchSegment^ CreateEllipticalArc(
    & System.double XC,
    & System.double YC,
    & System.double Zc,
    & System.double XMajor,
    & System.double YMajor,
    & System.double ZMajor,
    & System.double XMinor,
    & System.double YMinor,
    & System.double ZMinor,
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2,

```

& System.short [Direction](#)
)

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XC
X coordinate for the ellipse center point
YC
Y coordinate for the ellipse center point
Zc
Z coordinate for the ellipse center point
XMajor
X coordinate for a point on the ellipse and on the major axis
YMajor
Y coordinate for a point on the ellipse and on the major axis
ZMajor
Z coordinate for a point on the ellipse and on the major axis
XMinor
X coordinate for a point on the ellipse and on the minor axis
YMinor
Y coordinate for a point on the ellipse and on the minor axis
ZMinor
Z coordinate for a point on the ellipse and on the minor axis
X1
X coordinate for counter-clockwise elliptical arc start point
Y1
Y coordinate for counter-clockwise elliptical arc start point
Z1
Z coordinate for counter-clockwise elliptical arc start point
X2
X coordinate for counter-clockwise elliptical arc end point
Y2
Y coordinate for counter-clockwise elliptical arc end point
Z2
Z coordinate for counter-clockwise elliptical arc end point
Direction

- +1 : Go from the start point to the end point in a counter-clockwise direction
- 1 : Go from the start point to the end point in a clockwise direction

Return Value

[Sketch segment](#) for the elliptical arc

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateEllipticalArc](#).

Remarks

This method creates a full circle in the active 2D sketch. If a sketch is not active, then a new sketch is created. You can check for an active sketch using [ISketchManager::ActiveSketch](#).

[ISketchManager::AddToDB](#) and [ISketchManager::DisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database.

ISketchManager::AddToDB also avoids some of the peculiarities involved with creating entities via the user interface, such as inferencing, automatic relations, and snapping to the grid. Adding entities directly to the database also significantly increases the performance of this method. When you are done creating entities, it is important to ISketchManager::AddToDB(False), to restore SOLIDWORKS to its normal operating mode.

This method also works with ISketchManager::DisplayWhenAdded. If you have called ISketchManager::AddToDB(True), additional performance can be gained by calling ISketchManager::DisplayWhenAdded(False) to disable immediate display of entities as they are added to the database. When you are done creating all of your sketch entities, you must redraw your document window (see [IModelView::GraphicsRedraw](#) or [IModelView::IGraphicsRedraw](#)) to see the entities you added. You should also restore the original display settings by calling ISketchManager::DisplayWhenAdded(True).

To create a complete ellipse, use [ISketchManager::CreateEllipse](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateEquationSpline Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateEquationSpline Method (ISketchManager)

.NET Syntax

Visual Basic (Declaration)

```
Function CreateEquationSpline( _  
    ByVal YExpression As System.String, _  
    ByVal RangeStart As System.Double, _  
    ByVal RangeEnd As System.Double, _  
    ByVal IsAngleRange As System.Boolean, _  
    ByVal RotationAngle As System.Double, _  
    ByVal XOffset As System.Double, _  
    ByVal YOffset As System.Double, _  
    ByVal LockStart As System.Boolean, _  
    ByVal LockEnd As System.Boolean _  
) As SketchSpline
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim YExpression As System.String  
Dim RangeStart As System.Double  
Dim RangeEnd As System.Double  
Dim IsAngleRange As System.Boolean  
Dim RotationAngle As System.Double  
Dim XOffset As System.Double  
Dim YOffset As System.Double  
Dim LockStart As System.Boolean  
Dim LockEnd As System.Boolean  
Dim value As SketchSpline  
  
value = instance.CreateEquationSpline(YExpression, RangeStart, RangeEnd, IsAngleRange,  
RotationAngle, XOffset, YOffset, LockStart, LockEnd)
```

C#

```
SketchSpline CreateEquationSpline(  
    System.string YExpression,  
    System.double RangeStart,  
    System.double RangeEnd,  
    System.bool IsAngleRange,  
    System.double RotationAngle,  
    System.double XOffset,  
    System.double YOffset,  
    System.bool LockStart,  
    System.bool LockEnd  
)
```

C++/CLI

```
SketchSpline^ CreateEquationSpline(  
& System.String^ YExpression,  
& System.double RangeStart,  
& System.double RangeEnd,  
& System.bool IsAngleRange,  
& System.double RotationAngle,  
& System.double XOffset,  
& System.double YOffset,  
& System.bool LockStart,  
& System.bool LockEnd  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

YExpression
Equation for y (see **Remarks**)
RangeStart
Start value of x (see **Remarks**)
RangeEnd

End value of x (see **Remarks**)
IsAngleRange
True if the range and x value represent an angle (in radians), false if not
RotationAngle
Rotation angle (in radians) for the curve
XOffset
Offset in x for f(x), where x = 0 (see **Remarks**)
YOffset
Offset in y for f(x), where x = 0 (see **Remarks**)
LockStart
True to lock the start point (RangeStart) of the curve, false to not
LockEnd
True to lock the end point (RangeEnd) of the curve, false to not

Return Value

[Equation-driven curve](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateEquationSpline](#).

Example

[Create Equation-driven Curve \(C#\)](#)
[Create Equation-driven Curve \(VB.NET\)](#)
[Create Equation-driven Curve \(VBA\)](#)

Remarks

YExpression:

Any function that you can define as a 2D function in the following form:

`y-in-the-sketch = f(x-in-the-sketch)`

will appear as a curve in the sketch. For example:

`y = sin(x)`

RangeStart and RangeEnd

You cannot specify string values for RangeStart and RangeEnd; you must specify double values.

XOffset:

In the equation:

`y = sin(x) [x=0 to 2Pi]`

the start point (0,0) is moved by XOffset in the x direction.

YOffset:

In the equation:

`y = sin(x) [x=0 to 2Pi]`

the start point (0,0) is moved by YOffset in the y direction.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchSpline::GetEquationParameters Method](#)
[ISketchSpline::SetEquationParameters Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateEquationSpline2 Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateEquationSpline2 Method (ISketchManager)

Creates an equation-driven 2D explicit or parametric curve or a 3D parametric curve.

Visual Basic (Declaration)

```
Function CreateEquationSpline2( _
    ByVal XExpression As System.String, _
    ByVal YExpression As System.String, _
    ByVal ZExpression As System.String, _
    ByVal RangeStart As System.String, _
    ByVal RangeEnd As System.String, _
    ByVal IsAngleRange As System.Boolean, _
    ByVal RotationAngle As System.Double, _
    ByVal XOffset As System.Double, _
    ByVal YOffset As System.Double, _
    ByVal LockStart As System.Boolean, _
    ByVal LockEnd As System.Boolean _
) As SketchSpline
```

Visual Basic (Usage)



```
Dim instance As ISketchManager
Dim XExpression As System.String
Dim YExpression As System.String
Dim ZExpression As System.String
Dim RangeStart As System.String
Dim RangeEnd As System.String
Dim IsAngleRange As System.Boolean
Dim RotationAngle As System.Double
Dim XOffset As System.Double
Dim YOffset As System.Double
Dim LockStart As System.Boolean
Dim LockEnd As System.Boolean
Dim value As SketchSpline
```

```
value = instance.CreateEquationSpline2(XExpression, YExpression, ZExpression, RangeStart,
RangeEnd, IsAngleRange, RotationAngle, XOffset, YOffset, LockStart, LockEnd)
```

C#

```
SketchSpline CreateEquationSpline2(
    System.string XExpression,
    System.string YExpression,
    System.string ZExpression,
    System.string RangeStart,
    System.string RangeEnd,
    System.bool IsAngleRange,
    System.double RotationAngle,
    System.double XOffset,
    System.double YOffset,
    System.bool LockStart,
    System.bool LockEnd
)
```

C++/CLI

```
SketchSpline^ CreateEquationSpline2(
    & System.String^ XExpression,
    & System.String^ YExpression,
    & System.String^ ZExpression,
    & System.String^ RangeStart,
    & System.String^ RangeEnd,
    & System.bool IsAngleRange,
    & System.double RotationAngle,
    & System.double XOffset,
    & System.double YOffset,
    & System.bool LockStart,
    & System.bool LockEnd
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XExpression

For a parametric curve, equation for x in terms of t; for an explicit curve, an empty string (see **Remarks**)

YExpression

For a parametric curve, equation for y in terms of t; for an explicit curve, equation for y in terms of x (see **Remarks**)

ZExpression

Equation for z in terms of t (see **Remarks**)

RangeStart

Start value for x, if explicit; start value for t, if parametric (see **Remarks**)

RangeEnd

End value for x, if explicit; start value for t, if parametric (see **Remarks**)

IsAngleRange

True if the range and x value represent an angle (in radians), false if not (see **Remarks**)

RotationAngle

Rotation angle (in radians) for the curve (see **Remarks**)

XOffset

Translation in the x direction of the curve (see **Remarks**)

YOffset

Translation in the y direction of the curve (see **Remarks**)

LockStart

True to lock the start point (RangeStart) of the curve, false to not

LockEnd

True to lock the end point (RangeEnd) of the curve, false to not

Return Value

[Equation-driven curve](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateEquationSpline2](#).

Example

2D explicit curve:

A parabola $y = x^2$

CreateEquationSpline2("", "x^2", "", "-5", "5", False, 0, 0, 0, True, True)

2D parametric curve:

A spiral $x = t \sin(t)$; $y = t \cos(t)$

CreateEquationSpline2("t*sin(t)", "t*cos(t)", "", "0", "4*pi", False, 0, 0, 0, True, True)

3D parametric curve:

A helix $x = \sin(t)$; $y = \cos(t)$; $z = t/5$

CreateEquationSpline2("sin(t)", "cos(t)", "t/5", "0", "30", False, 0, 0, 0, True, True)

Example

[Create Equation-driven Curve \(C#\)](#)

[Create Equation-driven Curve \(VB.NET\)](#)

[Create Equation-driven Curve \(VBA\)](#)

Remarks

XExpression

If you want:

- an explicit curve (only in 2D) of the form $y=f(x)$, then this input should be "".
- a parametric curve given in terms of the parameter t, then this input is the value of x given in terms of the parameter. For example:

$x=\sin(t)$

YExpression

If you want:

- an explicit curve (only in 2D) of the form $y=f(x)$, then this input should be a function in terms of x to use to compute the value of y. For example:

$y=\exp(x)$

- a parametric curve given in terms of the parameter t, then this input is the value of y given in terms of the parameter. For example:

$y=\cos(t)$

ZExpression

For 3D parametric curves, this input is the value of z given in terms of parameter t. For example:

$z = t$

RangeStart and RangeEnd

Strings so that you can input dimension names or pi; for example, "10* width_dimension1" or "-pi/2".

XOffset, YOffset, IsAngleRange, and RotationAngle

In general, these parameters are not needed. They are for legacy support only. There are no equivalent 3D variables for positioning in 3D.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchSpline::GetEquationParameters2 Method](#)
[ISketchSpline::SetEquationParameters2 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateFillet Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateFillet Method (ISketchManager)

Creates a sketch fillet using the selected sketch entities.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateFillet( _  
    ByVal Radius As System.Double, _  
    ByVal ConstrainedCorners As System.Integer _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim Radius As System.Double  
Dim ConstrainedCorners As System.Integer  
Dim value As SketchSegment
```

```
value = instance.CreateFillet(Radius, ConstrainedCorners)
```

C#

```
SketchSegment CreateFillet(  
    System.double Radius,  
    System.int ConstrainedCorners  
)
```

C++/CLI

```
SketchSegment^ CreateFillet(  
& System.double Radius,  
& System.int ConstrainedCorners  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Radius

Radius of the fillet in meters

ConstrainedCorners

Action to take as defined in [swConstrainedCornerAction](#) e (see [Remarks](#))

Return Value

[Sketch segment](#) for the fillet

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateFillet](#).

Example

[Edit Radial Dimension \(C#\)](#)
[Edit Radial Dimension \(VB.NET\)](#)
[Edit Radial Dimension \(VBA\)](#)

Remarks

The ConstrainedCorners parameter indicates what action to take if the corner to fillet is constrained or dimensioned. If the corner is not constrained or dimensioned, then this parameter is ignored.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateLine Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateLine Method (ISketchManager)

Creates a sketch line in the currently active 2D or 3D sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateLine(  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double _  
) As SketchSegment
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X1 As System.Double  
Dim Y1 As System.Double  
Dim Z1 As System.Double  
Dim X2 As System.Double  
Dim Y2 As System.Double  
Dim Z2 As System.Double  
Dim value As SketchSegment
```

```
value = instance.CreateLine(X1, Y1, Z1, X2, Y2, Z2)
```

C#

```
SketchSegment CreateLine(  
    System.double X1,  
    System.double Y1,  
    System.double Z1,  
    System.double X2,  
    System.double Y2,
```

```
        System.double Z2  
    )
```

C++/CLI

```
SketchSegment^ CreateLine(  
&    System.double X1,  
&    System.double Y1,  
&    System.double Z1,  
&    System.double X2,  
&    System.double Y2,  
&    System.double Z2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X1](#)
X coordinate of the line start point
[Y1](#)
Y coordinate of the line start point
[Z1](#)
Z coordinate of the line start point
[X2](#)
X coordinate of the line end point
[Y2](#)
Y coordinate of the line end point
[Z2](#)
Z coordinate of the line end point

Return Value

[Sketch segment](#) for the line

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateLine](#).

Example

[Create Section View and Get Some Data \(VB.NET\)](#)
[Create Section View and Get Some Data \(VBA\)](#)
[Create Section View and Get Some Data \(C#\)](#)

Remarks

If a sketch is not active, then the line is not created and NULL is returned. You can check for an active sketch using the [ISketchManager::ActiveSketch](#) function.

[ISketchManager::AddToDB](#) and [ISketchManager::DisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database. ISketchManager::AddToDB also avoids inferencing.

When this method is used with a drawing document, this method creates the line relative to the active drawing view, [IDrawingDoc::ActiveDrawingView](#) or [IDrawingDoc::!ActiveDrawingView](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateLinearSketchStepAndRepeat Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateLinearSketchStepAndRepeat Method (ISketchManager)

Creates a linear sketch pattern.

.NET Syntax

```

Function CreateLinearSketchStepAndRepeat( _
    ByVal NumX As System.Integer, _
    ByVal NumY As System.Integer, _
    ByVal SpacingX As System.Double, _
    ByVal SpacingY As System.Double, _
    ByVal AngleX As System.Double, _
    ByVal AngleY As System.Double, _
    ByVal DeleteInstances As System.String, _
    ByVal XSpacingDim As System.Boolean, _
    ByVal YSpacingDim As System.Boolean, _
    ByVal AngleDim As System.Boolean, _
    ByVal CreateNumOfInstancesDimInXDir As System.Boolean, _
    ByVal CreateNumOfInstancesDimInYDir As System.Boolean _
) As System.Boolean

```

```

Dim instance As ISketchManager
Dim NumX As System.Integer
Dim NumY As System.Integer
Dim SpacingX As System.Double
Dim SpacingY As System.Double
Dim AngleX As System.Double
Dim AngleY As System.Double
Dim DeleteInstances As System.String
Dim XSpacingDim As System.Boolean
Dim YSpacingDim As System.Boolean
Dim AngleDim As System.Boolean
Dim CreateNumOfInstancesDimInXDir As System.Boolean
Dim CreateNumOfInstancesDimInYDir As System.Boolean
Dim value As System.Boolean

```

```

value = instance.CreateLinearSketchStepAndRepeat(NumX, NumY, SpacingX, SpacingY, AngleX, AngleY,
DeleteInstances, XSpacingDim, YSpacingDim, AngleDim, CreateNumOfInstancesDimInXDir,
CreateNumOfInstancesDimInYDir)

```

```

System.bool CreatelLinearSketchStepAndRepeat(
    System.int NumX,
    System.int NumY,
    System.double SpacingX,
    System.double SpacingY,
    System.double AngleX,
    System.double AngleY,
    System.string DeleteInstances,
    System.bool XSpacingDim,
    System.bool YSpacingDim,
    System.bool AngleDim,
    System.bool CreateNumOfInstancesDimInXDir,
    System.bool CreateNumOfInstancesDimInYDir
)

```

```

System.bool CreatelLinearSketchStepAndRepeat(
    & System.int NumX,
    & System.int NumY,
    & System.double SpacingX,
    & System.double SpacingY,
    & System.double AngleX,
    & System.double AngleY,
    & System.String^ DeleteInstances,
    & System.bool XSpacingDim,
    & System.bool YSpacingDim,
    & System.bool AngleDim,
    & System.bool CreateNumOfInstancesDimInXDir,

```

```
& System.bool CreateNumOfInstancesDimInYDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumX

Total number of instances along the x axis, including the seed

NumY

Total number of instances along the y axis, including the seed

SpacingX

Spacing between instances along the x axis

SpacingY

Spacing between instances along the y axis

AngleX

Angle for direction 1 relative to the x axis

AngleY

Angle for direction 2 relative to the y axis

DeleteInstances

Number of instances to delete, passed as a string in the format: "(a) (b) (c) "

XSpacingDim

True to display the spacing between instances dimension along the x axis in the graphics area, false to not

YSpacingDim

True to display the spacing between instances dimension along the y axis in the graphics area, false to not

AngleDim

True to display the angle dimension between axes in the graphics area, false to not

CreateNumOfInstancesDimInXDir

True to display the number of instances in the x direction dimension in the graphics area, false to not

CreateNumOfInstancesDimInYDir

True to display the number of instances in the y direction dimension in the graphics area, false to not

Return Value

True if the linear sketch pattern is created, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateLinearSketchStepAndRepeat](#).

Example

[Create and Edit Linear Sketch Pattern \(VB.NET\)](#)

[Create and Edit Linear Sketch Pattern \(VBA\)](#)

[Create and Edit Linear Sketch Pattern \(C#\)](#)

Remarks

The spacing instances, angle, and number of instances dimensions displayed in the graphics area can be modified by interactive users.

See Also

[ISketchManager Interface](#)

[ISketchManager Members](#)

[ISketchManager::EditLinearSketchStepAndRepeat Method](#)

[ISketchManager::CreateCircularSketchStepAndRepeat Method](#)

[ISketchManager::EditCircularSketchStepAndRepeat Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateParabola Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateParabola Method (ISketchManager)

Creates a parabola in the active sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateParabola( _  
    ByVal XFocus As System.Double, _
```

```

    ByVal YFocus As System.Double, _
    ByVal ZFocus As System.Double, _
    ByVal XApex As System.Double, _
    ByVal YApex As System.Double, _
    ByVal ZApex As System.Double, _
    ByVal X1 As System.Double, _
    ByVal Y1 As System.Double, _
    ByVal Z1 As System.Double, _
    ByVal X2 As System.Double, _
    ByVal Y2 As System.Double, _
    ByVal Z2 As System.Double _
) As SketchSegment

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim XFocus As System.Double
Dim YFocus As System.Double
Dim ZFocus As System.Double
Dim XApex As System.Double
Dim YApex As System.Double
Dim ZApex As System.Double
Dim X1 As System.Double
Dim Y1 As System.Double
Dim Z1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim Z2 As System.Double
Dim value As SketchSegment

```

```

value = instance.CreateParabola(XFocus, YFocus, ZFocus, XApex, YApex, ZApex, X1, Y1, Z1, X2, Y2, Z2)

```

C#

```

SketchSegment CreateParabola(
    System.double XFocus,
    System.double YFocus,
    System.double ZFocus,
    System.double XApex,
    System.double YApex,
    System.double ZApex,
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2
)

```

C++/CLI

```

SketchSegment^ CreateParabola(
    & System.double XFocus,
    & System.double YFocus,
    & System.double ZFocus,
    & System.double XApex,
    & System.double YApex,
    & System.double ZApex,
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XFocus
X coordinate for the focus
YFocus
Y coordinate for the focus
ZFocus
Z coordinate for the focus
XApex
X coordinate for the apex
YApex
Y coordinate for the apex
ZApex
Z coordinate for the apex
X1
X coordinate for the start point
Y1
Y coordinate for the start point
Z1
Z coordinate for the start point
X2
X coordinate for the end point
Y2
Y coordinate for the end point
Z2
Z coordinate for the end point

Return Value

[Sketch segment](#) for the parabola

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateParabola](#).

Example

[Create Parabola \(VBA\)](#)
[Create Parabola \(VB.NET\)](#)
[Create Parabola \(C#\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateConic Method \(\)](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateParallelogram Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateParallelogram Method (ISketchManager)

Creates a parallelogram.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateParallelogram( _  
    ByVal X1 As System.Double, _  
    ByVal Y1 As System.Double, _  
    ByVal Z1 As System.Double, _  
    ByVal X2 As System.Double, _  
    ByVal Y2 As System.Double, _  
    ByVal Z2 As System.Double, _  
    ByVal X3 As System.Double, _  
    ByVal Y3 As System.Double, _  
    ByVal Z3 As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X1 As System.Double
```

```

Dim Y1 As System.Double
Dim Z1 As System.Double
Dim X2 As System.Double
Dim Y2 As System.Double
Dim Z2 As System.Double
Dim X3 As System.Double
Dim Y3 As System.Double
Dim Z3 As System.Double
Dim value As System.Object

```

```
value = instance.CreateParallelogram(X1, Y1, Z1, X2, Y2, Z2, X3, Y3, Z3)
```

C#

```

System.Object CreateParallelogram(
    System.double X1,
    System.double Y1,
    System.double Z1,
    System.double X2,
    System.double Y2,
    System.double Z2,
    System.double X3,
    System.double Y3,
    System.double Z3
)

```

C++/CLI

```

System.Object^ CreateParallelogram(
    & System.double X1,
    & System.double Y1,
    & System.double Z1,
    & System.double X2,
    & System.double Y2,
    & System.double Z2,
    & System.double X3,
    & System.double Y3,
    & System.double Z3
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X1
X coordinate for point 1

Y1
coordinate for point 1
Z1
Z coordinate for point 1
X2
X coordinate for point 2
Y2
Y coordinate for point 2
Z2
Z coordinate for point 2
X3
X coordinate for point 3
Y3
Y coordinate for point 3
Z3
Z coordinate for point 3

Return Value

Array of [sketch segments](#) for the parallelogram

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateParallelogram](#).

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreatePoint Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreatePoint Method (ISketchManager)

Creates a sketch point in the active 2D or 3D sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePoint( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As SketchPoint
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As SketchPoint  
  
value = instance.CreatePoint(X, Y, Z)
```

C#

```
SketchPoint CreatePoint(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
SketchPoint^ CreatePoint(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X location of the point
Y
Y location of the point
Z
Z location of the point; ignored for 2D sketches

Return Value

Newly created [sketch point](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreatePoint](#).

Example

[Create Sketch Point \(VBA\)](#)
[Create Sketch Point \(VB.NET\)](#)
[Create Sketch Point \(C#\)](#)

Remarks

This method creates a point in the active 2D or 3D sketch. If a sketch is not active, the point is not created and NULL is returned. Use [ISketchManager.ActiveSketch](#) to check to see if the sketch is active.

[ISketchManager.AddToDB](#) and [ISketchManager.DisplayWhenAdded](#) increase performance during entity creation by adding entities directly to the SOLIDWORKS database. [ISketchManager.AddToDB](#) also avoids inferencing.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreatePolygon Method (ISketchManager)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreatePolygon Method (ISketchManager)

Creates a polygon in the active sketch.

.NET Syntax

Visual Basic (Declaration)

```
Function CreatePolygon( _  
    ByVal XC As System.Double, _  
    ByVal YC As System.Double, _  
    ByVal Zc As System.Double, _  
    ByVal Xp As System.Double, _  
    ByVal Yp As System.Double, _  
    ByVal Zp As System.Double, _  
    ByVal Sides As System.Integer, _  
    ByVal Inscribed As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchManager  
Dim XC As System.Double  
Dim YC As System.Double  
Dim Zc As System.Double  
Dim Xp As System.Double  
Dim Yp As System.Double  
Dim Zp As System.Double  
Dim Sides As System.Integer  
Dim Inscribed As System.Boolean  
Dim value As System.Object
```

```
value = instance.CreatePolygon(XC, YC, Zc, Xp, Yp, Zp, Sides, Inscribed)
```

C#

```
System.object CreatePolygon(  
    System.double XC,  
    System.double YC,  
    System.double Zc,  
    System.double Xp,  
    System.double Yp,  
    System.double Zp,  
    System.int Sides,  
    System.bool Inscribed  
)
```

```

System.Object^ CreatePolygon(
    & System.double XC,
    & System.double YC,
    & System.double Zc,
    & System.double Xp,
    & System.double Yp,
    & System.double Zp,
    & System.int Sides,
    & System.bool Inscribed
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[XC](#)
X coordinate for the center
[YC](#)
Y coordinate for the center
[Zc](#)
Z coordinate for the center
[Xp](#)
X coordinate of a vertex
[Yp](#)
Y coordinate of a vertex
[Zp](#)
Z coordinate of a vertex
[Sides](#)
Number of sides in the polygon
[Inscribed](#)
True to show an inscribed construction circle, false to show a circumscribed construction circle

Return Value

Array of [sketch segments](#) in the polygon

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreatePolygon](#).

Example

[Create Polygon \(VBA\)](#)
[Create Polygon \(VB.NET\)](#)
[Create Polygon \(C#\)](#)

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateRegionHatch Method (ISketchManager)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchManager Interface](#) : CreateRegionHatch Method (ISketchManager)

Creates an area hatch/fill region hatch using a closed sketch profile.

.NET Syntax

Visual Basic (Declaration)

```

Function CreateRegionHatch( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double, _
    ByVal Angle As System.Double, _
    ByVal Scale As System.Double, _
    ByVal Color As System.Integer, _

```

```

    ByVal Hatchname As System.String, _
    ByVal Layername As System.String _
) As SketchHatch

```

Visual Basic (Usage)



```

Dim instance As ISketchManager
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim Angle As System.Double
Dim Scale As System.Double
Dim Color As System.Integer
Dim Hatchname As System.String
Dim Layername As System.String
Dim value As SketchHatch

value = instance.CreateRegionHatch(X, Y, Z, Angle, Scale, Color, Hatchname, Layername)

```

C#

```

SketchHatch CreateRegionHatch(
    System.double X,
    System.double Y,
    System.double Z,
    System.double Angle,
    System.double Scale,
    System.int Color,
    System.string Hatchname,
    System.string Layername
)

```

C++/CLI

```

SketchHatch^ CreateRegionHatch(
    & System.double X,
    & System.double Y,
    & System.double Z,
    & System.double Angle,
    & System.double Scale,
    & System.int Color,
    & System.String^ Hatchname,
    & System.String^ Layername
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
X coordinate anywhere within the closed sketch profile

Y
Y coordinate anywhere within the closed sketch profile

Z
Z coordinate anywhere within the closed sketch profile

Angle
Angle of the hatch if the hatch is not solid

Scale
Scale factor for the hatch if the hatch is not solid

Color
COLORREF value describing the color used for the hatch

Hatchname
Type or name of the hatch from the **Sldwks.ptn** file

Layername
Layer name for the hatch if a drawing document

Return Value

[Sketch hatch](#)

Visual Basic for Applications (VBA) Syntax

See [SketchManager::CreateRegionHatch](#).

Remarks

This method creates a single area hatch/fill region hatch. All three point coordinates are important. This point is compared to all sketch contours to determine one unique external sketch region that contains the point.

See Also

[ISketchManager Interface](#)
[ISketchManager Members](#)
[ISketchManager::CreateBoundaryHatch Method](#)
[IModelDoc2::InsertHatchedFace Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0